

Midterm Review

June 16, 2026 6:23 PM

Mobile Applications Development – COMP 304

Midterm Practical Test Review

Test Date & Time: Tuesday, June 23, 2026 at 4:30 pm (2 hours)

Test Type: Hands-On Lab Practical (Android App Development)

- Open-book exam
- Students may use:
 - Lab computers or personal laptops
 - Android Studio IDE
 - Android Virtual Devices (AVD) / Emulators
 - Physical Android devices for testing and debugging
- Students must submit:
 - Android project files
 - Required screenshots demonstrating completed tasks
 - A compressed (.zip) submission file

Before the Exam

Ensure that the following are working properly:

- Android Studio is installed and updated
- Emulator or virtual device launches successfully
- Physical device debugging is enabled (if using a phone)
- Project creation and app execution can be completed without issues

Topics Covered

Students will be required to develop an Android application containing **two activities** and demonstrate the concepts covered in class.

1. Jetpack Compose User Interface

Be prepared to use and combine the following Compose UI components:

- Text
- TextField
- Button
- ImageButton
- RadioButton
- Checkbox
- Segmented Button / Segmented Control
- Image
- Other Material Design 3 components

Skills to Practice

- Creating reusable composable functions
- Organizing UI using Rows, Columns, and Boxes
- Aligning UI components properly
- Using LazyColumn for lists
- Applying Material Design 3 principles
- Building user-friendly interfaces

2. Android Architecture (MVVM)

Understand and implement:

Model

- Data classes
- Application data structures

View

- Compose UI screens
- User interface components

ViewModel

- Managing UI state
- Business logic
- Data communication between View and Model

3. Activity Navigation

Practice:

- Creating multiple activities
- Navigating between activities
- Passing data from Activity 1 to Activity 2
- Receiving and displaying passed data

4. User Interaction

Be able to:

- Handle button clicks
- Read TextField input
- Process RadioButton selections
- Process Checkbox selections
- Update UI based on user actions
- Display results dynamically

Coding Standards

Marks may be affected if coding standards are not followed.

Use Meaningful Names

Use clear and descriptive:

- Project names
- File names
- Variable names
- Function names
- Class names

Open-Book Exam Rules

Permitted Resources

Students may use:

- Course slides
- Textbooks
- Notes
- Assignments completed by themselves
- Programs created by themselves
- Solutions provided by the professor
- Android Developer Documentation
- Internet resources

Not Permitted

- AI tools that generate code
- External code assistants

Exception: Android Studio Gemini may be used if permitted by the instructor.

Additional Rules

- Students must use a PC or laptop.

- Students must submit their work through the Luminate Dropbox.
- Verify submission with the professor before leaving.
- Students may only use their own work or solutions provided by the professor.
- Sharing or using another student's solution is prohibited.

Common Reasons for Lost Marks

Marks may be deducted for:

1. Incorrect Naming

- Project name does not follow instructions
- Submission folder is incorrectly named

2. Missing Screenshots

- Required screenshots are not included

3. Reusing Existing Projects

- Students must create a new project for the exam
- Copying and modifying an old project is not permitted

4. Unresolved Errors

- Failure to troubleshoot and debug application issues

Submission Checklist

Before submitting:

- Project builds successfully
- App runs without errors
- All required screenshots are included
- Folder naming follows instructions
- ZIP file contains all necessary files

After submitting:

- Download the submitted ZIP file
- Verify that all files were uploaded correctly

- Confirm successful submission

Academic Integrity

All students must comply with Centennial College's Academic Integrity and Academic Honesty policies.

The following are considered violations:

- Plagiarism
- Cheating
- Unauthorized collaboration
- Using another student's work
- Submitting copied code

Submitted work may be reviewed for authenticity and originality using software tools.

Any academic misconduct will be handled according to Centennial College's Academic Integrity Policy and may result in disciplinary action.

Students are responsible for ensuring that all submitted work is their own.